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To: *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* editorial board.

Dear editors,

Please find attached our manuscript entitled "Time-series dimension reduction: a comprehensive review and conceptual unification of algorithms" that we are submitting for publication in *TPAMI*.

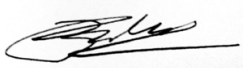
High-dimensional multivariate time-series data are ubiquitous in a range of fields, and dimension-reduction (DR) algorithms are important for the analysis of such data because they yield informative lower-dimensional representations which simplify computation and aid visualization and interpretation. Most DR methods that are used in practice— including principal component analysis (PCA) and autoencoder (AE) neural networks— are designed for general high-dimensional data, and thus do not account for the temporal structure of time-series data. However, there also exist many powerful but lesser-known DR methods that are sensitive to temporal permutation that we call time-series dimension reduction (TSDR) methods. Remarkably, a recent survey of DR techniques for time-series data (Ashraf et al., 2023, *IEEE Access*) **did not include *any* TSDR methods**, underscoring the underappreciation of this rich and powerful methodological literature.

A variety of powerful TSDR methods exist, developed across a range of disciplines, but they remain drastically underappreciated and have never previously been unified or reviewed.

The lack of awareness of TSDR methods has held back scientific progress involving high-dimensional time-series data (which more typically proceeds using general DR methods). Our work addresses these issues, by unifying the literature for the first time, and is positioned to become a key reference for future work on TSDR.

Our paper is of relevance to researchers in a range of areas including machine learning, time-series analysis, and complex systems. Accordingly, we expect our work to be of interest to the interdisciplinary readership of *TPAMI*. We propose that our paper best suits the *survey paper* type, since it includes a review and perspective on the field, in addition to original results and software. By providing an accessible unifying review of a drastically underappreciated methodological literature with an associated software package, we anticipate that this work will become a landmark paper, providing a foundational reference for the growing community empirical scientists analyzing high-dimensional time-series data. The framework it provides on the field and its existing inter-relationships further encourages the publication of exciting new theory and applications in *TPAMI*. Thank you for your consideration of our work.

Sincerely,



A/Prof. Ben D. Fulcher, on behalf of the authors